Electrochemistry

THE NERNST EQUATION

We know experimentally that the potential of a single electrode or half-cell varies with the concentration of ions in the cell. In 1889 Walter Nernst derived a mathematical relationship which enable us to calculate the half-cell potential, E, from the standard electrode potential, E°, and the temperature of the cell. This relation known as the Nernst equation can be stated as

$$E = E^\circ - \frac{2.303\,RT}{nF}\log K \qquad \text{...(1)}$$

where E° = standard electrode potential

R = gas constant

T = Kelvin temperature

n = number of electrons transferred in the half-reaction

F = Faraday of electricity

K = equilibrium constant for the half-cell reaction as in equilibrium law

Calculation of Half-cell potential

For an oxidation half-cell reaction when the metal electrode M gives M^{n+} ion,

$$M \longrightarrow M^{n+} + ne^-$$

the Nernst equation takes the form

$$E = E^\circ - \frac{2.303\,RT}{nF} \log \frac{[M^{n+}]}{[M]} \qquad ...(2)$$

The concentration of solid metal [M] is equal to zero. Therefore, the Nernst equation can be written as

$$E = E^\circ - \frac{2.303\,RT}{nF} \log [M^{n+}] \qquad ...(3)$$

Substituting the values of R, F and T at 25°C, the quantity 2.303 RT/F comes to be 0.0591. Thus the Nernst equation (3) can be written in its simplified form as

$$E = E^\circ - \frac{0.0591}{n} \log [M^{n+}]$$

This is the equation for a half-cell in which oxidation occurs. In case it is a reduction reaction, the sign of E will have to be reversed.

What is the potential of a half-cell consisting of zinc electrode in 0.01M ZnSO4 solution at 25°C, E° =0.763 V.

The half-cell reaction is

$$Zn \longrightarrow Zn^{2+} + 2e^-$$

The Nernst equation for the oxidation half-cell reaction is

$$E = E^\circ - \frac{0.0591}{n}\log[Zn^{2+}]$$

The number of electrons transferred $n = 2$ and E° =0.763 V.

Substituting these values in the Nernst equation we have

$$E = 0.763 - \frac{0.0591}{2}\log(0.01)$$

$$= 0.763 - \frac{0.0591}{2}(-2)$$

$$= 0.763 + 0.0591 = \mathbf{0.8221\ V}$$

Calculation of Cell potential

The Nernst equation is applicable to cell potentials as well. Thus,

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log K$$

K is the equilibrium constant of the redox cell reaction.

SOLVED PROBLEM. Calculate the emf of the cell.

$$Zn \mid Zn^{2+}(0.001M) \parallel Ag^{+}(0.1M) \mid Ag$$

The standard potential of Ag/Ag^{+} half-cell is + 0.80 V and Zn/Zn^{2+} is – 0.76 V.

Step 1. Write the half-cell reactions of the anode and the cathode. Then add the anode and cathode half reactions to obtain the cell reaction and the value of E°_{cell}.

Cathode :	$2Ag^{+} + 2e^{-} \longrightarrow 2Ag$	$E^{\circ} = +0.80\ V$
Anode :	$Zn \longrightarrow Zn^{2+} + 2e^{-}$	$E^{\circ} = -0.76\ V$
Cell :	$Zn + 2Ag^{+} \rightleftharpoons Zn^{2+} + 2Ag$	$E^{\circ} = 1.56\ V$

Step 2. K for the cell reaction $= \dfrac{[Zn^{2+}]}{[Ag^{+}]^{2}}$

substitute the given values in the Nernst equation and solving for E_{cell}, we have

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log K$$

$$= 1.56 - \frac{0.0591}{2} \log \frac{[Zn^{2+}]}{[Ag^{+}]^{2}}$$

$$= 1.56 - \frac{0.0591}{2} \log \frac{[10^{-3}]}{[10^{-1}]^{2}}$$

$$= 1.56 - 0.02955\,(\log 10^{-1})$$

$$= 1.56 + 0.02955$$

Calculation of Equilibrium constant for the cell reaction

The Nernst equation for a cell is

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log K$$

At equilibrium, the cell reaction is balanced and the potential is zero. The Nernst equation may, now, be written as

$$0 = E^\circ_{cell} - \frac{0.0591}{n} \log K$$

or

$$\log K = \frac{n E^\circ_{cell}}{0.0591}$$

SOLVED PROBLEM. Calculate the equilibrium constant for the reaction between silver nitrate and metallic zinc.

SOLUTION

Step 1. Write the equation for the reaction

$$2Ag^+ + Zn \rightleftharpoons Zn^{2+} + 2Ag \qquad E^\circ cell = 1.56\,V$$

Step 2. Substitute values in the Nernst equation at equilibrium

$$\log K = \frac{nE^\circ_{cell}}{0.0591}$$

$$0 = 1.56 - 0.03 \log K$$

$$-1.56 = -0.03 \log K$$

$$\log K = \frac{-1.56}{-0.03} = 52$$

$$K = \mathbf{1 \times 10^{52}}$$